

DINO-MOT: 3D Multi-Object Tracking With Visual Foundation Model for Pedestrian Re-Identification Using Visual Memory Mechanism

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Abstract—In the advancing domain of autonomous driving, this research focuses on enhancing 3D Multi-Object Tracking (3D-MOT). Pedestrians are particularly vulnerable in urban environments, and robust tracking methodologies are required to understand their movements. Prevalent Tracking-By-Detection (TBD) frameworks often underutilize the rich visual data from sensors such as cameras. This study leverages the advanced visual foundation model, DINOv2, to refine the TBD framework by incorporating camera modality, thereby improving pedestrian tracking consistency and overall 3D-MOT performance. The proposed DINO-MOT framework is the first application of DINOv2 for enhancing 3D-MOT through pedestrian Re-Identification (Re-ID), and Score Filter Ceiling is implemented to prevent premature exclusion of low-confidence 3D detections during tracking association. Furthermore, utilization of DINOv2 as a feature extractor within the DINO-MOT framework reduces pedestrian ID switches by up to 12.3%. Achieving AMOTA of 76.3% on the nuScenes test dataset, DINO-MOT has set a new benchmark in the 3D MOT literature with an improvement of 0.5%, securing the top rank on the leaderboard. Furthermore, this research paves the potential of applying a visual foundation model to improve the existing TBD framework, to enhance 3D-MOT in autonomous driving.

Index Terms—Intelligent transportation systems, human detection and tracking, deep learning for visual perception.

I. INTRODUCTION

PERCEPTION is a fundamental task within the domain of autonomous driving, playing a crucial role in accurately detecting and interpreting the behaviors exhibited by various road-users. Downstream tasks, such as planning and control, rely heavily on the capabilities of detection and tracking, necessitating high-performance levels [1]. Considerable research efforts are being directed towards 3D object detection and tracking, which are critical in advancing autonomous vehicles towards practical applications. Pedestrians, among the road users, are disproportionately more susceptible to fatal risks in urban environments, being ten times more likely to encounter fatality compared to vehicle occupants [2]. In order to enhance pedestrian

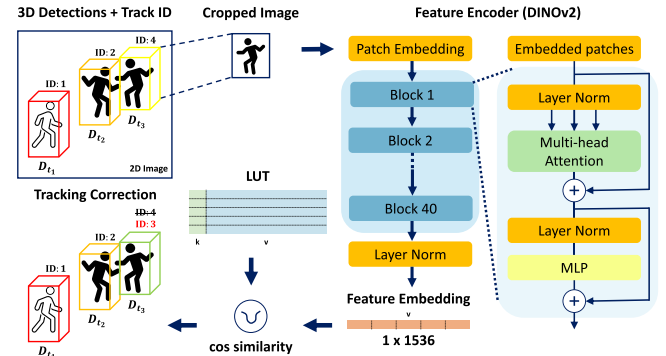


Fig. 1. **Overview of Pedestrian Re-Identification using Visual Foundation Model.** Given 3D detection D_t from any sensor modality with corresponding track ID T_t , the detections are projected onto a 2D image plane. Pedestrian images with detection confidence scores exceeding 0.7 are extracted and fed into DINOv2 to obtain their visual features $f_{ped} \in \mathbb{R}^{1 \times 1536}$. These features are stored in a Lookup Table (LUT) as key-value pairs, where the pedestrian feature vector is the value and the corresponding track ID is the key. The stored feature vector f_{ped} undergoes a cosine similarity check against existing pedestrian feature vectors in the LUT. If a track ID stored in the LUT has the highest similarity score that surpasses the similarity threshold, the current pedestrian's track ID is updated with the stored track ID to perform pedestrian tracking correction.

safety, it is crucial to monitor and understand their movements. However, pedestrian tracking presents significant challenges due to variations in scale, potential motion blur, and the complexities introduced by crowded environments, which can lead to overlapping and occlusion among pedestrians [3]. Current leading algorithms for tracking utilize the Tracking-By-Detection (TBD) framework [4], which leverages initial detections to facilitate the tracking process. However, the framework underutilizes the potential of sensors used in 3D detections. For example, camera images offer rich visual information that could value-add to the tracking process, yet this data is often discarded after post-detection in the TBD framework. With the emergence of advanced visual foundation models, this research aims to refine the TBD framework by integrating camera modality, thereby enhancing the consistency of pedestrian tracking and overall 3D-MOT performance. In DINO-MOT, DINOv2 [5] is utilized to enhance pedestrian tracking within the context of 3D-MOT, as depicted in Fig. 1. The contributions of this research are as follows:

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- It introduces the first application of a visual foundation model, specifically DINOv2, to enhance 3D-MOT through pedestrian Re-Identification (Re-ID).
- A Score Filter Ceiling is implemented to mitigate the premature exclusion of 3D detections with low confidence during the initial phase of tracking association.
- DINO-MOT, a 3D Multi-Object Tracking framework, is proposed, which employs DINOv2 as a feature extractor to assist in Pedestrian Re-ID, effectively reducing pedestrian ID switches by up to 12.3%.
- DINO-MOT has demonstrated state-of-the-art performance on the nuScenes test benchmark, registering Average Multi-Object Tracking Accuracy (AMOTA) of 76.3%, to accomplish the first rank on the leaderboard.

In the following sections, the related work and the proposed methodology are discussed, followed by the experimental result of DINO-MOT to benchmark it against existing methods.

II. RELATED WORK

A. 3D Multi-Object Tracking (3D-MOT)

3D Multi-Object Tracking methods can be classified based on their modality. For instance, using 3D lidar, ShaSTA [6] learns shape and spatiotemporal affinities between consecutive frames to minimize false positives and negatives, while Zaech et al. [7] employ a graph-based learning approach to represent detections and tracks, though these methods often neglect texture and appearance information. On the other hand, EagerMOT [8] leverages on both 2D and 3D detectors for long-distance tracking without additional training, while DeepFusionMOT [9] balances speed and accuracy by updating 2D trajectories with 3D lidar data when distant objects are detected solely by cameras. Also, some methods use various metrics to improve tracking performance. P3DMMOT [10] introduces a learnable metric that combines Mahalanobis and feature distances in data association, while PF-MOT [11] proposes using Euclidean distance and feature similarity for its tracking metrics. Moreover, recent advancements in 3D-MOT predominantly employ the Tracking-By-Detection (TBD) framework [12]. CAMO-MOT [13] boosts accuracy with a motion cost matrix and multi-category loss in intricate environments. While Poly-MOT [14] refines the TBD framework through the implementation of diverse motion models and a two-stage data association process, Fast-Poly [4] leverages on rotated object alignment, and parallel computation techniques to improve the speed of the tracking. Based on AMOTA, Fast-Poly (75.8%), Poly-MOT (75.4%), CAMO-MOT (75.3%) are the leading 3D-MOT algorithms on nuScenes benchmark [15]. Notably, the leading 3D-MOT algorithms follow TBD frameworks, and they often underutilize the sensors employed for 3D detections, which could be used to improve the consistency in tracking of objects in occluded scenarios, or crowded environments. Hence, this study aims to explore the integration of cameras to improve 3D-MOT performance within the TBD framework.

B. Pedestrian Re-Identification (Re-ID)

Pedestrian Re-ID refers to recognizing a previously seen pedestrian, and assigning it the existing track ID. The initial predominance of CNNs set the stage for significant advancements in pedestrian Re-ID [16]. Subsequently, the introduction of transformer-based methods marked a revolutionary shift, consistently outperforming CNN-based strategies [17], but requiring significant computational resources. The pioneering TransReID [18] was instrumental in integrating Vision Transformer (ViT) into Re-ID, opening new avenues for exploration. Also, the development of a pyramidal transformer structure mirrors the multi-scale feature learning typical of CNNs, enhancing the patch embedding process [19]. Recently, the emergence of pre-trained large-scale foundation models offers a new opportunity to apply these general models to specialized Re-ID tasks. The integration of the visual foundation model with ViT has demonstrated remarkable performance enhancements in pedestrian Re-ID [16]. This success has spurred the development of specialized models like PASS [20], which elevate pedestrian Re-ID capabilities further. These transformative advances underscore the potential of applying visual foundation models in pedestrian Re-ID.

III. PROPOSED SOLUTIONS

DINO-MOT, a 3D Multi-Object Tracking algorithm is introduced, inspired by Poly-MOT [14]. It leverages on the visual understanding capabilities of DINOv2 [5] for pedestrian Re-ID, leading to enhanced tracking performance. An overview of pedestrian Re-ID through the implementation of visual memory and DINOv2 is depicted in Fig. 1. Moreover, an overview of DINO-MOT is shown in Fig. 2.

A. Pedestrian Image Extraction

In accordance with the TBD framework, 3D detections (D_t) and their respective track IDs (T_t) from the current frame t are used as inputs. Each pedestrian's 3D detection is projected onto a 2D image (camera view). Firstly, 3D detections from the world coordinates are transformed to camera coordinates as shown in (1) [15]. $D_{t,camera}$, D_t refer to 3D detections in camera coordinates, and world coordinates respectively, while $T_{ego \rightarrow camera}$, $T_{world \rightarrow ego}$ refer to transformation matrix from ego-vehicle to camera and from world to ego-vehicle respectively. Subsequently, 8 corners of $D_{t,camera}$ are acquired. Given one of the corners, $p_{camera} = [x, y, z]^T$, homogeneous coordinate $p_{hom} = [x, y, z, 1]^T$ is obtained. Then, transform the point into homogeneous coordinate p_{proj} using camera intrinsic matrix K as shown in (2) [15], where f_x , f_y are focal lengths in x and y coordinates, c_x , c_y are x and y coordinates of principal point. Afterwards, project the point onto a 2D image p_{img} using (3) [15], resulting in 2D camera pixel coordinates $[u, v]$.

$$D_{t,camera} = T_{ego \rightarrow camera} \cdot T_{world \rightarrow ego} \cdot D_t \quad (1)$$

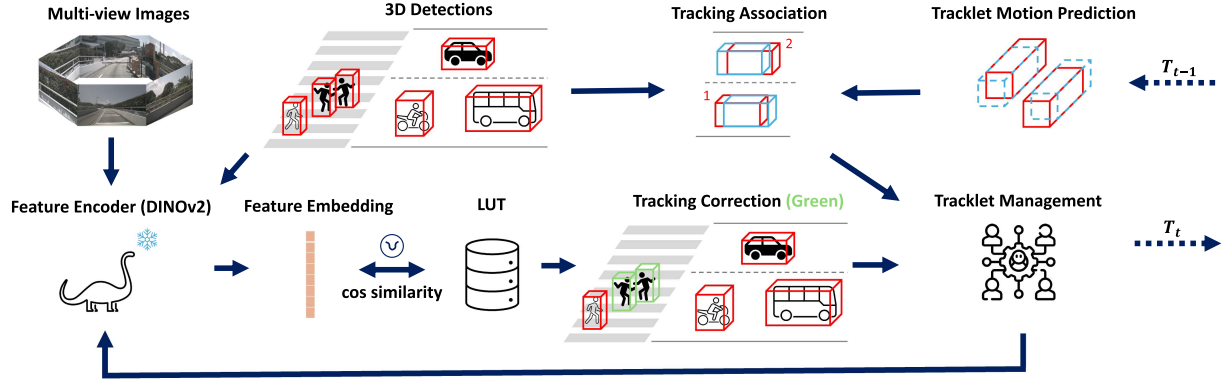


Fig. 2. **Pipeline of DINO-MOT at frame t .** 3D object detections are refined using Score Filter, retaining those with confidence above it to reduce false positives. A Score Filter Ceiling is introduced to prevent excessive exclusion of low-confidence detections. Subsequently, NMS is applied, yielding pre-processed detections D_t . Tracklets from the previous frame T_{t-1} undergo motion prediction using CTRA or bicycle models to account for class-specific behaviors. The Extended Kalman Filter (EKF) is applied to estimate trajectory states to handle non-linear movements. Two stage tracking association matches the predicted state from the previous frame $T_{t,t-1}$ with D_t using the Hungarian algorithm. The tracklet management module updates trajectories using EKF, initiates new tracklets from unmatched detections, and terminates tracklets exceeding their lifespan. Given multi-view images with D_t , the 3D detections are projected onto 2D image. Pedestrian image I is extracted, and processed by DINOv2 to obtain its visual feature f_{ped} . f_{ped} is stored in a Lookup Table (LUT) with its track ID in the current frame T_t and checked against existing features in the LUT using cosine similarity. If an existing track ID in the LUT has the highest similarity score exceeding the similarity threshold, it overwrites the current pedestrian's track ID for tracking correction (pedestrian Re-ID). The corrected track ID is updated in the tracklet management module, finalizing the tracking results T_t for subsequent frames.

$$\mathbf{P}_{proj} = K \cdot \mathbf{P}_{hom} = \begin{bmatrix} f_x X + c_x Z \\ f_y Y + c_y Z \\ Z \end{bmatrix} \quad (2)$$

$$\mathbf{P}_{img} = \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} \frac{f_x X + c_x Z}{f_y Y + c_y Z} \\ \frac{f_y Y + c_y Z}{Z} \end{bmatrix} = \begin{bmatrix} \frac{f_x X}{Z} + c_x \\ \frac{f_y Y}{Z} + c_y \end{bmatrix} \quad (3)$$

With the eight corners obtained in 2D image perspective, find the minimum and maximum value of x , y coordinates amongst them respectively to obtain x_{min} , x_{max} , y_{min} , y_{max} . Hence, the four corners representing the region of interest is defined as $PROI = [(x_{min}, y_{min}), (x_{max}, y_{min}), (x_{min}, y_{max}), (x_{max}, y_{max})]$, with its area encapsulating all the vertices of $D_{t,camera}$. Pedestrian images I , are extracted from the region of interest before being processed by the Visual Feature Encoder. The quality of the extracted pedestrian image is closely associated with the 3D detection confidence score (C_{ped}). Therefore, only 3D detections with confidence scores exceeding 0.7 are used for pedestrian image extraction. Fig. 3 shows a comparison of pedestrians extracted at various confidence scores. Some pedestrians with confidence score less than 0.4 have occlusion from the surrounding structures, and some are captured at a far distance from the camera which caused them to have a low resolution. From the visualization, pedestrians with a confidence score beyond 0.7 shows a more clear view with less occlusion. Thus, C_{ped} is defined as 0.7.

B. Visual Feature Encoder

The visual feature encoder transforms each pedestrian image I into a high-level feature vector representation $f_{ped} \in \mathbb{R}^{1 \times 1536}$, to apply it for re-identification. This study employs DINO-v2, specifically its largest variant, DINOv2-giant, without additional training or fine-tuning. Hence, all the weights of DINOv2 remain

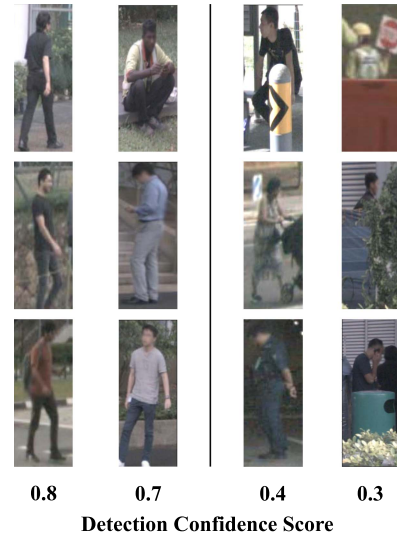


Fig. 3. Visual Comparison of Extracted Pedestrian Images in Various Confidence Score on nuScenes dataset.

frozen in this study. The process involves passing cropped pedestrian images through a patch-embedded module composed of a 2D convolutional layer with a kernel size of 14×14 , with a stride of 14, followed by a normalization layer. The uniform output patches of size 14×14 are processed by 40 Vision Transformer (ViT) blocks into a feature vector, as depicted in Fig. 1 and (4).

$$f_{ped} = \text{DINO}(I) \quad (4)$$

C. Visual Memory

A Lookup Table (LUT) stores the visual feature vectors f_{ped} along with their corresponding track IDs, serving as a visual memory database. As the LUT is designed to handle increasing data volumes efficiently, it implements memory management

strategies by retaining pedestrian features for a maximum duration of 20 seconds before purging. As shown in Fig. 1, the LUT is represented as a set of key-value pairs in (5), where $f_{\text{ped}_i} \in \mathbb{R}^{1 \times 1536}$ refers to pedestrian feature vectors (value), $\text{ID}_i \in \mathbb{Z}$ refers to respective track ID of the pedestrian (key).

$$\text{LUT} = \{(\text{ID}_1, f_{\text{ped}_1}), (\text{ID}_2, f_{\text{ped}_2}), \dots, (\text{ID}_N, f_{\text{ped}_N})\} \quad (5)$$

D. Pedestrian Re-Identification (Re-ID)

Upon the detection of a new pedestrian, its visual feature vector $f_{\text{ped}_{\text{new}}}$ is compared against all the stored feature vectors in the LUT f_{ped_i} using cosine similarity according to the (6), where size of N is fixed at feature size 1536 as $f_{\text{ped}} \in \mathbb{R}^{1 \times 1536}$.

$$\text{CS}(f_{\text{ped}_{\text{new}}}, f_{\text{ped}_i}) = \frac{\sum_{j=1}^N (f_{\text{ped}_{\text{new},j}} \cdot f_{\text{ped}_{i,j}})}{\sqrt{\sum_{j=1}^N f_{\text{ped}_{\text{new},j}}^2} \cdot \sqrt{\sum_{j=1}^N f_{\text{ped}_{i,j}}^2}} \quad (6)$$

A stringent similarity threshold of 0.975 is set empirically to determine if two pedestrian track IDs are similar. The existing track ID in the LUT with the highest similarity score that surpasses this threshold are a match. The old track ID stored in the LUT will overwrite the current track ID and the new visual feature vector will overwrite the old visual feature vector, a process known as pedestrian tracking correction or Re-ID. If similarity matching fails, the visual feature vectors and their track IDs are stored in the LUT.

E. 3D Object Detection

DINO-MOT incorporates IS-FUSION [21] as its 3D object detector, leveraging both lidar and camera inputs. With an image resolution of 384×1056 and 10 lidar sweeps per frame, IS-FUSION is selected for its superior detection performance on the nuScenes benchmark. Notably, 3D object detector with any sensor modality can be selected. In pre-processing of the 3D object detections, Score Filter is applied to retain only those detections with confidence scores exceeding the set threshold, thereby reducing false-positive detections [14]. Understanding that there is a potential for true positive detections among low-confidence detections, Score Filter Ceiling (SFC) is implemented to prevent excessive exclusion. Subsequently, Non-Maximum Suppression (NMS)[22] is performed, resulting in pre-processed detections at frame t (D_t). The Score Filter thresholds are empirically defined as $\{0.012, 0.048, 0.044, 0.014, 0.030, 0.018, 0.026\}$ for Bicycle, Bus, Car, Motorcycle, Pedestrian, Trailer, Truck respectively, with a Score Filter Ceiling set at 0.05. The experiments are detailed in Section IV.

F. Tracklet Motion Prediction

In order to achieve robust 3D Multi-Object Tracking, it necessitates the consideration of highly nonlinear motion patterns exhibited by various road users. The choice of an appropriate motion model is critical, given the distinct motion characteristics of different object classes. The Constant Turn Rate and Acceleration (CTRA) model assumes uniform angular velocity and constant acceleration. It is suitable for Car, Bus, Truck, Trailer,

and Pedestrian movements [14]. Conversely, the Bicycle model, which focuses on steering angle and velocity, is ideal for Bicycle and Motorcycle movements. In order to address the nonlinear nature of the object class behavior, the Extended Kalman Filter (EKF) is implemented to estimate the trajectory state.

G. Tracking Association

The effectiveness of tracking association varies by object class due to its unique geometric characteristics, hence suitable similarity metrics and association thresholds vary as well. Generalized Intersection Over Union (GIOU) is used as the similarity metric in both Bird's Eye View (BEV) and 3D as shown in (7), (8), where B refers to 3D bounding box of detection D_t or tracklet $T_{t,t-1}$, C refers to the convex hull of B_i and B_j .

$$\text{GIOU}_{3D}(B_i, B_j) = \frac{|B_i \cap B_j|_{3D}}{|B_i \cup B_j|_{3D}} - \frac{|C - (B_i \cup B_j)|_{3D}}{|C|_{3D}} \quad (7)$$

$$\text{GIOU}_{BEV}(B_i, B_j) = \frac{|B_i \cap B_j|_{BEV}}{|B_i \cup B_j|_{BEV}} - \frac{|C - (B_i \cup B_j)|_{BEV}}{|C|_{BEV}} \quad (8)$$

Two-stage tracking association is implemented to reduce false negatives by considering more detections and tracklets, by applying both GIOU methods. If the first stage association uses GIOU-BEV, the second stage employs GIOU-3D, and vice versa, ensuring both similarity metrics are utilized in the tracking association process. In the first stage, GIOU-BEV is applied for bus and GIOU-3D for other classes. Upon obtaining $T_{t,t-1}$ (predicted state at $t-1$) and D_t , the first cost matrix C_1 is constructed according to the respective class similarity metrics. $C_1 = \{C_{1bic}, C_{1bus}, C_{1car}, C_{1motor}, C_{1ped}, C_{1tra}, C_{1tru}\}$, where the acronyms represent Bicycle, Motorcycle, Pedestrian, Trailer, and Truck, respectively. Each object class within C_1 is aggregated prior to applying the Hungarian algorithm, matching D_t and $T_{t,t-1}$ based on C_1 . Accounting for the geometric size difference among object classes and to constrain the first stage tracking association, the first association threshold is empirically defined as $\{1.55, 1.34, 1.31, 1.44, 1.69, 1.25, 1.21\}$ for Bicycle, Bus, Car, Motorcycle, Pedestrian, Trailer, and Truck respectively. Subsequently, unmatched detections and tracklets proceed to a second stage of tracking association, utilizing the alternate similarity metrics to form the second cost matrix C_2 . The Hungarian algorithm is again applied with a standardized second association threshold of 1.0. The process culminates in the identification of final matched pairs, unmatched detections, and unmatched tracklets.

H. Tracklet Management

The tracklet management module operates through continuous updating of trajectories using the EKF to predict the states of matched tracklets. This module initiates new tracklets from detections that remain unmatched and terminates tracklets that surpass a maximum lifespan of 10 seconds. This ensures the 3D-MOT framework to maintain a responsive management of tracklets.

TABLE I

ABLATION STUDY OF DINO-MOT ON NUSCENES VALIDATION SPLIT. IMPROVED PERFORMANCE IS MARKED IN **BOLD**. (BIC, MOTOR, PED, TRA, TRU) REFERS TO (BICYCLE, MOTORCYCLE, PEDESTRIAN, TRAILER, TRUCK)

Detector	DINO	AMOTA (%) $\uparrow$	IDS $\downarrow$							
			Overall	Bic	Bus	Car	Motor	Ped	Tra	Tru
CenterPoint [23]	No	73.1	229	2	0	76	3	143	1	4
CenterPoint [23]	Yes	73.1	216	2	0	76	3	130	1	4
CMT [24]	No	77.4	343	5	2	99	8	219	3	7
CMT [24]	Yes	77.4	316	5	2	99	8	192	3	7
IS-Fusion [21]	No	80.3	288	2	3	121	6	146	5	5
IS-Fusion [21]	Yes	80.3	274	2	3	121	6	132	5	5

TABLE II

COMPARISON OF EXISTING 3D OBJECT TRACKING ALGORITHMS ON NUSCENES TEST SPLIT. THE BEST PERFORMANCE IS MARKED IN **BOLD**. (BIC, MOTOR, PED, TRA, TRU) REFERS TO (BICYCLE, MOTORCYCLE, PEDESTRIAN, TRAILER, TRUCK)

Method	AMOTA (%) $\uparrow$								AMOTP (<i>m</i>) $\downarrow$	IDS $\downarrow$	FP $\downarrow$	FN $\downarrow$
	Overall	Bic	Bus	Car	Motor	Ped	Tra	Tru				
BEVFusion [25]	74.1	56.0	77.9	83.1	74.8	83.7	73.4	69.5	0.403	506	19997	19395
NEBP-V3 [26]	74.6	49.9	78.6	86.1	80.7	83.4	76.1	67.3	0.498	377	17243	19316
CAMO-MOT [13]	75.3	59.2	77.7	85.8	78.2	85.8	72.3	67.7	0.472	324	17269	18192
Poly-MOT [14]	75.4	58.2	78.6	86.5	81.0	82.0	75.1	66.2	0.422	292	19673	17956
Fast-Poly [4]	75.8	57.3	76.7	86.3	82.6	85.2	76.8	65.6	0.479	326	17098	18415
DINO-MOT (<i>ours</i>)	76.3	56.5	78.4	85.9	78.5	86.7	75.8	72.5	0.520	387	16017	17066

TABLE III

ANALYSIS ON AMOTA OF DINO-MOT USING VARIOUS SF VALUES ON NUSCENES VALIDATION SPLIT. THE BEST PERFORMANCE IS MARKED IN **BOLD**. (SF, BIC, MOTOR, PED, TRA, TRU) REFERS TO (SCORE FILTER, BICYCLE, MOTORCYCLE, PEDESTRIAN, TRAILER, TRUCK)

SF	AMOTA (%) $\uparrow$						
	0.03	0.05	0.07	0.09	0.11	0.13	0.15
Bic	71.29	69.77	68.38	68.08	66.39	66.45	64.46
Bus	88.58	88.64	88.57	86.30	86.30	86.28	83.84
Car	88.66	88.63	86.88	86.94	84.91	84.93	84.95
Motor	83.75	83.80	81.63	81.54	79.54	79.67	77.44
Ped	86.62	86.76	85.17	85.24	85.36	83.41	83.50
Tra	62.25	61.05	59.20	59.72	58.36	58.18	56.70
Tru	76.61	76.41	74.94	75.11	73.32	73.45	71.60
Overall	79.68	79.29	77.83	77.56	76.31	76.05	74.64

IV. EXPERIMENTS

A. Dataset

nuScenes dataset [15] is a well-known benchmark for research in autonomous driving, featuring data collected from multiple perception sensors which includes careful calibration of sensor intrinsic, extrinsic parameters, including six cameras, one lidar, and five radars. This dataset consists of methodically divided 700 train, 150 validation, and 150 test scenes. The evaluation of tracking methodologies within this dataset adheres to the official nuScenes 3D object tracking metrics: Average Multi-Object Tracking Accuracy (AMOTA), Average Multi-Object Tracking Precision (AMOTP), ID Switches (IDS), False Positives (FP), and False Negatives (FN). These metrics form a robust and standardized framework essential for the assessment and validation of 3D Multi-Object Tracking performance in autonomous driving scenarios.

B. Score Filter Ceiling

Score Filter Ceiling is defined based on its impact on AMOTA. As demonstrated in Table III, experiments are conducted on the nuScenes dataset and the highest AMOTA across

all classes is obtained when the score filter is set at 0.05 or lower. While using a higher Score Filter excludes more low-confidence detection boxes, thereby reducing the overall operational time and making it more favorable for real-time autonomous driving application, a Score Filter Ceiling of 0.05 is identified as optimal for AMOTA performance according to the experiments. Similarly, an empirical analysis is conducted with a higher precision to determine the most appropriate Score Filter for each class, as defined in Section III-E.

C. Ablation Study

In order to assess the impact of the DINO-MOT framework, a series of comparative experiments are conducted, both with and without the integration of DINO. Experiments with DINO include the pedestrian Re-ID, whilst experiments without DINO have the pedestrian Re-ID module deactivated. The results of these comparative studies are presented in Table I. As a proof of concept, DINO-MOT is implemented alongside three distinct 3D object detection algorithms: CenterPoint [23], CMT [24], and IS-Fusion [21]. The findings indicate a consistent reduction in pedestrian ID Switch (IDS) across all three detectors. This reduction is attributed to DINO-MOT's targeted object class re-identification mechanism aimed at minimizing IDS for the specific object class, in this case, pedestrians. The results clearly show a decline in pedestrian IDS without impacting other object classes. Specifically, the decline in pedestrian IDS is 13 for CenterPoint, 27 for CMT, and 14 for IS-Fusion, translating that there is up to 12.3% reduction in pedestrian IDS. These results demonstrate that DINO-MOT effectively reduces pedestrian IDS, thereby enhancing the consistency of 3D Multi-Object Tracking. Moreover, it is noteworthy that the AMOTA metrics remain unaffected across all three detectors, suggesting that the pedestrian Re-ID is a non-invasive method that does not compromise tracking accuracy.



Fig. 4. Qualitative comparison of DINO-MOT and Poly-MOT [14], on nuScenes dataset. Both algorithms use IS-Fusion [21] as a 3D detector.

D. Official Benchmark

DINO-MOT is benchmarked on the well-known nuScenes dataset, to contextualize its performance against existing 3D Multi-Object Tracking algorithms, as shown in Table II. The official benchmark reveals that DINO-MOT has achieved the highest AMOTA of 76.3%, establishing itself as the new state-of-the-art. DINO-MOT surpasses the previous leading algorithm, Fast-Poly [4], with a 0.5% improvement in AMOTA, a 6.3% reduction in False Positives (FP), and a 7.3% reduction in False Negatives (FN). This indicates that DINO-MOT delivers outperforming tracking performance while reducing FP, which lowers the likelihood of mismatched tracking associations. Moreover, DINO-MOT has lower FN, minimizing the cost of overlooking objects, which is a critical factor for autonomous driving applications. Specifically, DINO-MOT achieved state-of-the-art AMOTA of 86.7% for pedestrians and 72.5% for trucks across the benchmark, indicating its superior tracking accuracy in these categories. Overall, DINO-MOT has set a new benchmark in 3D Multi-Object Tracking, firmly establishing itself as the leading solution on the nuScenes benchmark.

E. Visualization

The tracking performance of DINO-MOT is qualitatively compared against Poly-MOT [14] in various scenarios, particularly focusing on its effectiveness in reducing unnecessary pedestrian IDS. Fig. 4 illustrates the visual comparisons of DINO-MOT and Poly-MOT, from two different sequences in the nuScenes dataset. Both algorithms use IS-Fusion [21] as a 3D detector. Also, each track ID is represented by a unique color of the bounding box. The first sequence highlights the challenge of pedestrian IDS in an occluded and crowded environment. It depicts a bus stop in NUS, where a man's track ID highlighted by a red arrow is shown from frame 4599 to frame 4606. For Poly-MOT, the track ID of the pedestrian changes across the frames due to occlusion by a road sign pole, demonstrating an ID switch. Conversely, the track ID remains consistent for DINO-MOT, indicating the effectiveness of the Re-ID feature in DINO-MOT. Moreover, the second sequence demonstrates the contribution of DINO-MOT to consistent track ID assignment. There is a noticeable change in pedestrian track ID from frame

TABLE IV
ANALYSIS OF DINO-MOT COMPUTATIONAL TIME

DINOv2 Variant	Computational Overhead (ms)	
	DINOv2	Cosine Similarity
Small	2.54	0.067
Base	2.57	0.106
Large	4.41	0.124
Giant	7.24	0.207

5148 to 5156 for Poly-MOT, unlike DINO-MOT. These visual comparisons underscore the utility of DINOv2 in DINO-MOT for reducing pedestrian IDS occurrence. Also, it provides insight into its role in enhancing overall tracking performance in autonomous driving applications.

F. Discussion

1) *Runtime*: The run time of DINO-MOT is analyzed, with particular focus on its two primary components: DINOv2 inference and the cosine similarity operation. In this experiment, all DINOv2 model variants (small, base, large, and giant) are evaluated, as demonstrated in Table IV.

Both the inference and the corresponding cosine similarity computation time are measured to quantify the additional computational overhead introduced by incorporating DINOv2 for pedestrian re-identification in DINO-MOT. The experiments are conducted using Nvidia RTX 3090Ti GPU for DINOv2 inference and Intel i9-12900 K CPU for cosine similarity computations. The average computational overhead for each pedestrian is determined based on the processing of 1,000 pedestrian images from the nuScenes dataset. The DINOv2-giant variant achieves an average inference time of 7.24 ms (138 FPS), while the cosine similarity operation for pedestrian features takes 0.207 ms on average (4,831 FPS). These results suggest that DINO-MOT, leveraging the DINOv2-giant variant, is well-suited for real-time autonomous driving applications. Additionally, employing smaller DINOv2 variants reduces the computational overhead for both inference and cosine similarity operations, potentially enhancing the overall time efficiency of DINO-MOT.

2) *Expanding DINO-MOT*: DINO-MOT selects pedestrians as the primary class for re-identification, introducing a novel approach that leverages visual foundation models to enhance

overall 3D multi-object tracking performance in autonomous driving applications. Pedestrians exhibit the highest incidence of tracking ID switches (IDS) compared to other object classes, as shown in Table I. This elevated IDS is primarily attributed to the complexities of densely populated environments, where factors such as occlusion and overlapping significantly impact tracking accuracy [3]. Additionally, pedestrians are particularly more vulnerable to fatal risks in urban settings, in contrast to vehicle occupants [2]. As a result, the pedestrian class is prioritized for re-identification within the DINO-MOT framework. Future work will expand DINO-MOT's capabilities by incorporating visual memory mechanisms for other classes, including vehicles and cyclists, as these are equally vital for improving comprehensive tracking performance in the context of autonomous driving.

V. CONCLUSION

This research explores 3D Multi-Object Tracking in autonomous driving. With the aid of the visual foundation model DINOv2, the study aims to reduce unnecessary pedestrian track ID switches (IDS) by re-identifying pedestrians based on its visual features, using a visual memory mechanism. Moreover, Score Filter Ceiling is implemented to prevent over-exclusion of low-confidence 3D detections. On top of that, DINO-MOT has been introduced, demonstrating a reduction in pedestrian IDS by up to 12.3%. Furthermore, it has achieved AMOTA of 76.3% on the nuScenes test dataset, establishing a new benchmark in the literature. Future research will focus on extending the scope of DINO-MOT and enhancing tracking association for other object classes - such as vehicles and cyclists, to improve the overall tracking performance.

REFERENCES

- [1] J. Mao, S. Shi, X. Wang, and H. Li, "3D object detection for autonomous driving: A comprehensive survey," *Int. J. Comput. Vis.*, Springer, no. 8, pp. 1909–1963, 2023.
- [2] A. Soathong, D. Wilson, P. Ranjitkar, and S. Chowdhury, "A critical review of policies on pedestrian safety and a case study of New Zealand," *Sustainability*, vol. 11, no. 19, 2019, Art. no. 5274.
- [3] U. Gawande, K. Hajari, and Y. Golhar, "Real-time deep learning approach for pedestrian detection and suspicious activity recognition," *Procedia Comput. Sci.*, vol. 218, pp. 2438–2447, 2023.
- [4] X. Li, D. Liu, Y. Wu, X. Wu, L. Zhao, and J. Gao, "Fast-poly: A fast polyhedral framework for 3D multi-object tracking," *IEEE Robot. Automat. Lett.*, vol. 9, no. 11, pp. 10519–10526, 2024.
- [5] M. Oquab et al., "Dinov2: Learning robust visual features without supervision," *Trans. Mach. Learn. Res.*, 2024.
- [6] T. Sadjadpour, J. Li, R. Ambrus, and J. Bohg, "ShaSTA: Modeling shape and spatio-temporal affinities for 3D multi-object tracking," *IEEE Robot. Automat. Lett.*, vol. 9, no. 5, pp. 4273–4280, May 2024.
- [7] J.-N. Zaech, A. Liniger, D. Dai, M. Danelljan, and L. Van Gool, "Learnable online graph representations for 3D multi-object tracking," *IEEE Robot. Automat. Lett.*, vol. 7, no. 2, pp. 5103–5110, Apr. 2022.
- [8] A. Kim, A. Ošep, and L. Leal-Taixé, "EagerMOT: 3D multi-object tracking via sensor fusion," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2021, pp. 11315–11321.
- [9] X. Wang, C. Fu, Z. Li, Y. Lai, and J. He, "DeepFusionMOT: A 3D multi-object tracking framework based on camera-LiDAR fusion with deep association," *IEEE Robot. Automat. Lett.*, vol. 7, no. 3, pp. 8260–8267, Jul. 2022.
- [10] H.-k. Chiu, J. Li, R. Ambrus, and J. Bohg, "Probabilistic 3D multi-modal, multi-object tracking for autonomous driving," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2021, pp. 14227–14233.
- [11] T. Wen, Y. Zhang, and N. M. Freris, "PF-MOT: Probability fusion based 3D multi-object tracking for autonomous vehicles," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2022, pp. 700–706.
- [12] P. Zhang, X. Li, L. He, and X. Lin, "3D multiple object tracking on autonomous driving: A literature review," 2023, *arXiv:2309.15411*.
- [13] L. Wang et al., "CAMO-MOT: Combined appearance-motion optimization for 3D multi-object tracking with camera-LiDAR fusion," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 11, pp. 11981–11996, Nov. 2023.
- [14] X. Li et al., "Poly-MOT: A polyhedral framework for 3D multi-object tracking," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2023, pp. 9391–9398.
- [15] H. Caesar et al., "nuScenes: A multimodal dataset for autonomous driving," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 11621–11631.
- [16] M. Ye, S. Chen, C. Li, W.-S. Zheng, D. Crandall, and B. Du, "Transformer for object re-identification: A survey," *Int. J. Comput. Vis.*, 2024.
- [17] B. Zhang, Y. Liang, and M. Du, "Interlaced perception for person re-identification based on swin transformer," in *Proc. IEEE 7th Int. Conf. Image. Vis. Comput.*, 2022, pp. 24–30.
- [18] S. He, H. Luo, P. Wang, F. Wang, H. Li, and W. Jiang, "TransReID: Transformer-based object re-identification," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 15013–15022.
- [19] H. Li, M. Ye, C. Wang, and B. Du, "Pyramidal transformer with conv-patchify for person re-identification," in *Proc. 30th ACM Int. Conf. Multimedia*, 2022, pp. 7317–7326.
- [20] K. Zhu, H. Guo, T. Yan, Y. Zhu, J. Wang, and M. Tang, "PASS: Part-aware self-supervised pre-training for person re-identification," in *Proc. Eur. Conf. Comput. Vis.*, Springer, 2022, pp. 198–214.
- [21] J. Yin et al., "Is-Fusion: Instance-scene collaborative fusion for multimodal 3D object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 14905–14915.
- [22] J. Hosang, R. Benenson, and B. Schiele, "Learning non-maximum suppression," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 4507–4515.
- [23] T. Yin, X. Zhou, and P. Krahenbuhl, "Center-based 3D object detection and tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 11784–11793.
- [24] J. Yan et al., "Cross modal transformer: Towards fast and robust 3D object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 18268–18278.
- [25] Z. Liu et al., "BEVFusion: Multi-task multi-sensor fusion with unified bird's-eye view representation," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2023, pp. 2774–2781.
- [26] M. Liang and F. Meyer, "Neural enhanced belief propagation for multiobject tracking," *IEEE Trans. Signal Process.*, vol. 72, pp. 15–30, 2024.